

Théo Matricon

Optimisation AI Scientist — Python — Search Algorithms — Machine Learning

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Summary

Computer Science PhD and ML/software researcher focused on **algorithmic optimization, search, constraints, and code generation**.

Experienced in formulating hard combinatorial problems, building **Python/C++** research systems, and benchmarking algorithms at cluster scale for **solver-ready, production-oriented implementations**.

Technical Skills

Optimization / OR: exact search, branch-and-bound style enumeration, constraints, objective functions, cost models, benchmarking

Programming / ML / Systems: Python, C/C++, CUDA C++, Cython, Java, SQL, PyTorch, LLMs, Linux, Git, compute clusters

Experience

Researcher

INRIA, Rennes, France

Postdoctoral Researcher – Algorithmic Optimization, Code Generation & AI

January 2025 – Present

- **Modeling:** Formulate synthesis and configuration problems as objective-driven search tasks with explicit candidate spaces, constraints, costs, and evaluation criteria.
- **Optimization:** Developed algorithms that reduced large-scale benchmarking compute costs by **50% on average** while preserving reliable performance comparisons.
- **Solver-style Delivery:** Build reproducible Python pipelines to configure, run, and compare optimisation approaches across datasets, baselines, and hardware scales.
- **HPC:** Deployed and evaluated software on 50+ nodes clusters using **200k+ CPU hours**.
- **Technical Leadership:** Supervise 3 interns, advise 5 junior researchers, and present results to mixed research and engineering audiences.

Junior Researcher

University of Bordeaux / LaBRI, Bordeaux, France

PhD Researcher – Search Algorithms & Code Generation

October 2021 – December 2024

- **Search Algorithms:** Developed a constant-delay best-first search algorithm that ran **2× faster** than prior methods and **scaled exponentially better** with search-space complexity.
- **Optimization Framework:** Designed and maintained DeepSynth2 (20,000+ lines of code), an experimental framework for implementing, benchmarking, and comparing search-based optimisation algorithms.
- **Hybrid Methods:** Combined learned models, symbolic constraints, and external knowledge to solve problems beyond conventional synthesizers; related techniques were subsequently applied to **code generation for Excel**.
- **Development:** Implemented performance-critical algorithms in **Python, C/C++, Cython, and PyTorch**.
- **Communication:** Published **10+ peer-reviewed papers**, including at leading AI venues such as AAAI, and presented at international conferences and workshops.

Education

PhD in Computer Science

University of Bordeaux, France

Program Synthesis and Code Generation

2021 – 2024

Engineering Degree

ENSEIRB-MATMECA, Bordeaux, France

Engineering Degree / Master's in Computer Science – Artificial Intelligence

2018 – 2021